

Definition

$$\mathbb{S}^N = \text{span}\{e^{ik \cdot x} \mid k \in \mathcal{N}^d\} \cap \mathbb{R}$$

$$\mathbb{S}_+^N = \{f_N \in \mathbb{S}^N \mid f_N(x_k) \geq 0, \forall x_k = 2\pi k / (2N + 1), k \in \mathcal{N}^d\}.$$

Lemma

Making use of the discrete Fourier transform and Parseval's theorem, it is not difficult to see that for any two functions g and w in $\mathbb{S}^N$,

$$\langle g \mid w \rangle = (2\pi)^d \sum_{k \in \mathcal{N}^d} \hat{g}_k \overline{\hat{w}_k} = \frac{(2\pi)^d}{(2N + 1)^d} \sum_{k \in \mathcal{N}^d} g(x_k) w(x_k).$$

Statement

I try to prove that: the following projection

$$\Pi_+^N f := \underset{g \in \mathbb{S}_+^N}{\text{argmin}} \|g - f\|_2^2, \quad \text{s.t.} \quad \int_{-\pi}^{\pi} g dx = \int_{-\pi}^{\pi} f dx.$$

is equivalent to

$$P_+^N f := \underset{g \in \mathbb{S}_+^N}{\text{argmin}} \|g - P^N f\|_2^2, \quad \text{s.t.} \quad \frac{(2\pi)^d}{(2N + 1)^d} \sum_{k=-N}^N g(x_k) = \int_{-\pi}^{\pi} f dx.$$

where P^N is the L^2 projection. Since both g is in $\mathbb{S}^N$, by Parseval's theorem,

$$\begin{aligned} \|g - f\|_2^2 &= \|P^N(g - f)\|_2^2 + \|g - f - P^N(g - f)\|_2^2 \\ &= \|g - P^N f\|_2^2 + \|f - P^N f\|_2^2 \end{aligned}$$

As the minimization is taken on g , $\|f - P^N f\|_2^2$ is just a given constant which can be removed from argmin . The first equivalent is shown. Then, by the lemma, as g and $P^N f$ are in $\mathbb{S}^N$, it holds that

$$\|g - P^N f\|_2^2 = \frac{(2\pi)^d}{(2N + 1)^d} \sum_{k \in \mathcal{N}^d} (g(x_k) - P^N f(x_k))^2.$$

Now you can see, everything is taken at collocation points. In particular, when f itself is in $\mathbb{S}^N$, we can remove the operator P^N as $P^N f = f$.

Equivalence of W_h and $\mathbb{S}_+^N \cap \{f \mid \int_{-\pi}^{\pi} f dx = 1\}$

Without loss of generality, I modified the definition of W_h in your paper, to make the constant consistent.

$$W_h = \{\vec{\rho}_h = (\rho_{-N}, \dots, \rho_N)^T \mid \frac{(2\pi)^d}{(2N+1)^d} \|\vec{\rho}\|_{l^1} = 1, \vec{\rho} \geq 0\}.$$

For any $f \in \mathbb{S}_+^N \cap \{f \mid \int_{-\pi}^{\pi} f dx = 1\}$, it is obvious that $\vec{f} \geq 0$ from the definition of $\mathbb{S}_+^N$. By the lemma,

$$1 = \int_{-\pi}^{\pi} f dx = \langle f \mid 1 \rangle = \frac{(2\pi)^d}{(2N+1)^d} \sum_{k \in \mathcal{N}^d} f(x_k) = \frac{(2\pi)^d}{(2N+1)^d} \|\vec{f}\|_{l^1},$$

which shows $f \in W_h$. Vice versa.